



cost of security. With zero configuration, there's typically no password, no encryption, and open ports, making the device highly vulnerable. It's akin to purchasing a laptop and connecting it to the internet without any personalisation or security measures in place. This ease of setup often leads to significant security vulnerabilities in IoT devices," he explains.

Cleaning the lens

While Verma's statement allows us to heave a sigh of relief, it also means that surveillance camera vulnerabilities will persist, but negligence can trigger drastic outcomes, which rogue users can hack. The lack of a dedicated government authority to formulate and implement cybersecurity policies and regulations in India has resulted in a glaring absence of comprehensive safety and security rules that should have been established and communicated to the general populace. "This lack of attention is perplexing, especially in the context of India's *Digital India* initiative," Dr Sunila remarks.

"If there were a dedicated cybersecurity ministry, it could have enforced policies mandating that all imported devices undergo vulnerability assessments and penetration testing. These tests would ascertain the presence of vulnerabilities, and only devices deemed secure would receive certification. However, this system is currently absent, and there is no official certification or licensing process for vendors supplying these devices," Sunila adds.

In contrast, countries with dedicated cybersecurity ministries often establish guidelines for specific sectors. Hospitals, schools, and other institutions are provided with clear recommendations regarding the quality and specifications of CCTV systems they should employ. However, such guidelines are absent in India, resulting in consumers making blind choices based on advertisements

without a clear understanding of the technical aspects involved.

Verma places the responsibility for securing the system on the user. "When you buy a laptop, it's your responsibility to set a secure password. In the case of security cameras, the data in question often belongs to the user or the organisation using the cameras. This data includes sensitive and personal information, making it the user's responsibility to safeguard it," he says.

While the internet holds the potential to poison a system, it also provides the antidote. An online search engine called Shodan (Sentient Hyper-Optimised Data Access Network), launched by John Matherly in 2009, maps and gathers information about internet-connected devices and systems, including surveillance cameras. Verma explains how his team had leveraged the data from Shodan and other search engines to warn users of possible vulnerabilities in the equipment. "Even if you don't have control over the device firmware, our platform equips the user with tools that test these devices for vulnerabilities such as exposure to man-in-the-middle attacks, weak passwords, screen capture vulnerabilities, and the security of video streams, providing insights into the security status of the devices."

The pill, however effective it may be, needs to be taken regularly to avoid the chronic disease that threatens security, but all of us have heard about how prevention is better than cure! With the self-reliance in manufacturing mission undertaken by the government, the answer seems pretty clear—Make (it safe) in India!

Subramanhya and his team have developed an edge-processing autonomous IoT device consisting of multiple sensors, including dual cameras, speakers and microphones, motion sensors, and GPS, using Intel's

OpenVINO toolkit. "Developing the source code for camera systems is a lengthy and intricate process, as it involves understanding the complexities of optics, colour rendering, analogue-to-digital circuits, and scene illumination, among other factors. Creating a high-quality camera requires a deep knowledge of various technologies. After six years, we have successfully achieved that feat," he declares.

The scarcity of talent and skill has never been a question. Dr Sunila believes that many individuals in India possess the technical knowledge required to understand and assess the security of these devices. "However, when it comes to installing and monitoring CCTV systems, two critical factors must be considered: professional installation and monitoring. While individuals can be trained for self-monitoring in home settings, organisations should opt for professional monitoring. Establishing an incident response team and a Security Operations Centre (SOC) should also be part of the security framework," she adds.

Security cameras were initially installed to create a secure environment for individuals and citizens. Ironically, modern technology has enabled them to become a threat to security. Weak passwords, insufficient safeguards, human negligence and inability are the perfect conditions for creating gateways for rogue elements. There is a paramount need for a competent certification body that ensures devices entering the country are thoroughly tested, certified, and, if necessary, vulnerabilities are rectified before they reach consumers. While building an industry that is self-reliant in surveillance equipment manufacturing may take years, the matter at hand is time-sensitive and grows graver with the looming threat of an Orwellian future. **EFY**

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